Homework 5

Due Thursday, October 8, 2026 by 11:59 PM on Gradescope

Instructions

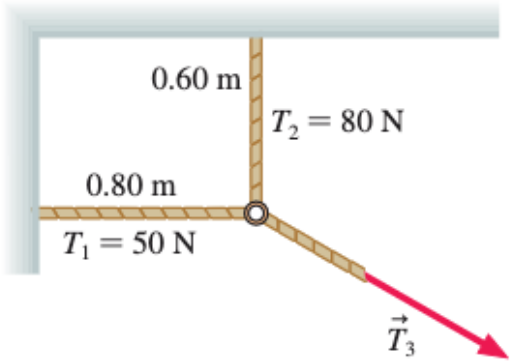
- Include your full name and Coyote ID on the top right of the first page
- Clearly circle or box your answer for any question involving calculations. If the problem is multipart, circle or box answers for each part.
- Work should be neatly written and easy to follow.
- You must tag each question on Gradescope. If you do not tag your work for a given problem, you will not receive credit for it.

Problems

This homework contains problems related forces, Newton's second law, mechanical equilibrium, constant forces, and friction.

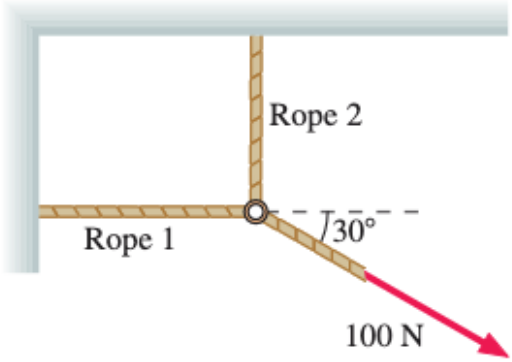
Mechanical equilibrium model

FIGURE 1



1. The three ropes in Figure 1 are tied to a small, very light ring (i.e., its size and weight does not contribute to the problem). Two of these ropes are anchored to walls at right angles with the tensions shown in the figure. What are the magnitude and direction of the tension $\vec{T}_3$ in the third rope?

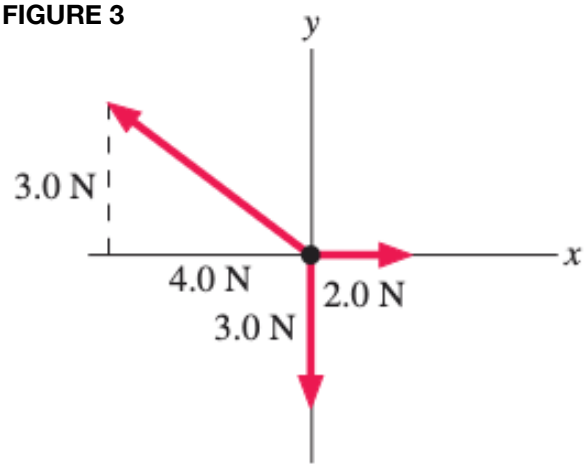
FIGURE 2



2. The three ropes in Figure 2 are tied to a small, very light, ring (i.e., its size and weight does not contribute to the problem). Two of the ropes are anchored to walls at right angles, and the third rope pulls as shown. What are T_1 and T_2, the magnitudes of the tension forces in the first two ropes?
3. A football coach sits on a sled while two of this players build their strength by dragging the sled across the field with ropes. The friction force on the sled is $1000\,\mathrm{N}$, the players have equal pulls, and the angle between the two ropes is 20°. How hard must each player pull to drag the coach at a steady $2.0\,\mathrm{m/s}$?

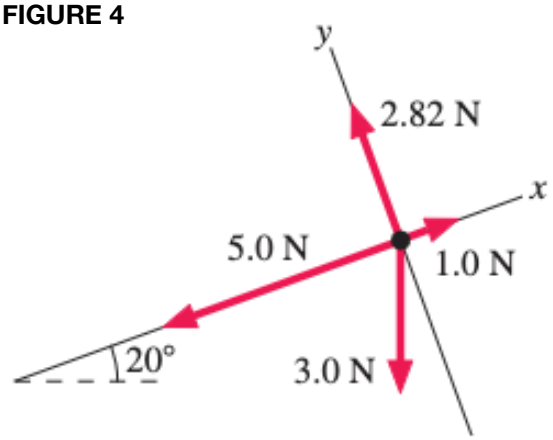
Constant force model

FIGURE 3



4. The forces in Figure 3 act on a $2.0\,\text{kg}$ object. What are the values of a_x and a_y

FIGURE 4



5. The forces in Figure 4 act on a $2.0\,\text{kg}$ object. What are the values of a_x and a_y?

6. A $55\,\mathrm{kg}$ water skier is pulled by a horizontal tow rope. She goes from rest to $6.0\,\mathrm{m/s}$ in $4.5\,\mathrm{s}$ when the rope's tension is $120\,\mathrm{N}$. What is the magnitude of the friction force between her skis and the water?

Friction

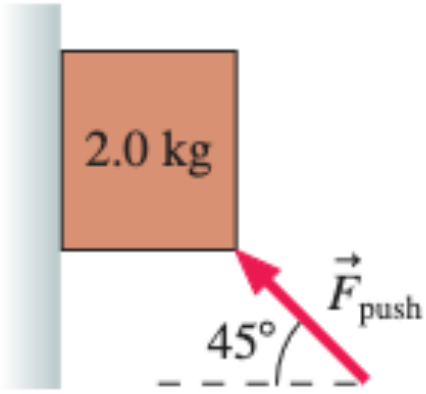
7. Bonnie an dClyde are sliding a $300\,\mathrm{kg}$ bank safe across the floor to their getaway car. The safe slides with a constant speed (i.e., mechanical equilibrium) if Clyde pushes from behind with $385\,\mathrm{N}$ of force while Bonnie pulls forward on a rope with $350\,\mathrm{N}$ of force. What is the safe's coefficient of kinetic friction on the bank floor?
8. Bob is pulling a $30\,\mathrm{kg}$ filing cabinet with a force of $200\,\mathrm{N}$, but the filing cabinet refuses to move (i.e., static friction and mechanical equilibrium). The coefficient of static friction between the filing cabinet and the floor is 0.80. What is the magnitude of the friction force on the filing cabinet?
9. A $4000\,\mathrm{kg}$ truck is parked on a 15° slope (i.e., static friction and mechanical equilibrium). How big is the friction force on the truck? The coefficient of static friction between the tires and the road is 0.90.

Problems

10. Henry, whose mass is $95\,\mathrm{kg}$, stands on a bathroom scale in an elevator. The scale reads $830\,\mathrm{N}$ for the first $3.0\,\mathrm{s}$ after the elevator starts moving, then $930\,\mathrm{N}$ for the next $3.0\,\mathrm{s}$.What is the elevator's velocity $6.0\,\mathrm{s}$ after starting? (Hint: the value that Henry reads is their net force, the only other force you have consider is Henry's weight)
11. A truck with a heavy load has a total mass of $7500\,\mathrm{kg}$. It is climbing a 15° incline at a steady $15\,\mathrm{m/s}$ when, unfortunately, the poorly secured load falls off! Immediately after losing the load, the truck begins to accelerate at $1.5\,\mathrm{m/s^2}$. What was the mass of the load? Ignore rolling friction.
12. An object of mass m is at rest at the top of a smooth slope of height h and length L. The coefficient of kinetic friction between the object and the surface, μ_k, is small enough that the object will slide down the slope after being given a very small push to get it started. Find an expression for the object's speed at the bottom of the slope.
13. Sam, whose mass is $75\,\mathrm{kg}$, takes off across level snow on his jet-powered skis. The skis have a trust of $200\,\mathrm{N}$ and a coefficient of kinetic friction on snow of 0.10. Unfortunately, the skis run out of fuel after only $10\,\mathrm{s}$.
    a. What is Sam's top speed?
    b. How far has Sam traveled when he finally coasts to a stop?

14. A large box of mass M is pulled across a horizontal, friction-less surface by a horizontal rope with tension T. A small box of mass m sits on top of the large box. The coefficient of static and kinetic friction between the two boxes are μ_s and μ_k, respectively. Find an expression for the maximum tension $T_{\max}$ for which the small box rides on top the large box without slipping.

FIGURE 5



15. The $2.0\,\text{kg}$ food box in Figure 5 slides down a vertical wood wall while you push on it at a $45°$ angle. What magnitude of force should you apply to cause the box to slide down at constant speed? (Use $\mu_k = 0.2$)